

Improving Autonomous Robot Localization using GPS and IMU Data Fusion: Progress Report 3

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Abstract—This report covers the Week 3 progress of the GPS/IMU sensor fusion project. Static noise characterization measurements were conducted for both the MPU-6050 IMU and the u-blox 7 GPS receiver. The resulting bias and standard-deviation values quantify sensor noise and serve as the measurement-noise covariance inputs for the upcoming Extended Kalman Filter. A notable gyroscope bias on the Y-axis was identified, which requires explicit bias compensation in the filter design.

Index Terms—GPS, IMU, Sensor Fusion, Kalman Filter, Noise Characterization

I. INTRODUCTION

This project fuses GPS and IMU data on an RC-car platform to improve localization accuracy beyond what a standalone u-blox 7 GPS receiver provides [1], [2]. Progress Reports 1 and 2 covered hardware bring-up, IMU driver development, and the first ROS2 sensor nodes. Week 3 focused on quantifying the noise characteristics of both sensors, a prerequisite for setting the EKF covariance matrices.

II. PROGRESS TO DATE

Static recordings were taken with the vehicle stationary outdoors. IMU data (/imu/data_raw, 100 Hz) was captured for 90 s; the first 5 s were discarded to exclude initialization transients. GPS fixes (/gps/fix, 1 Hz) were recorded for 5.5 min (345 valid fixes). All analysis was performed in a Jupyter notebook using the rosbags library [3].

IMU noise characterization: Fig. 1 shows the six raw channels over the recording window. Bias (mean at rest) and noise standard deviation were computed for each axis and are given in Table I. The accelerometer Z-axis reads 9.654 m/s², consistent with gravity at ± 4 g full-scale. The most significant finding is a gyroscope Y-axis bias of 0.105 rad/s (≈ 6.0 °/s). Left uncorrected during integration, this bias would accumulate to a 360° heading error in under one minute. The EKF must therefore either subtract the measured bias before fusion or include it as an estimated state variable.

GPS accuracy baseline: Fig. 2 shows the 2-D position scatter and altitude time series. With the receiver stationary, 50 % of fixes fall within 1.84 m of the mean position (CEP50) and 95 % within 3.49 m (CEP95), both within the 2.5–5 m specification of the u-blox 7 [1]. The position standard deviations (1.60 m North, 1.26 m East) will be used directly as the diagonal entries of the GPS measurement-noise covariance matrix R .

EKF covariance initialization: Both noise covariances follow from the measurements: $R_{\text{GPS}} = \text{diag}(1.60^2, 1.26^2)$ m² from the position scatter; R_{IMU} will be finalized together with the state vector in Week 4. For the Gyro Y bias, two approaches are considered: subtracting the measured offset as a pre-filter correction, or augmenting the state vector with an estimated bias. The choice will be made during Week 4.

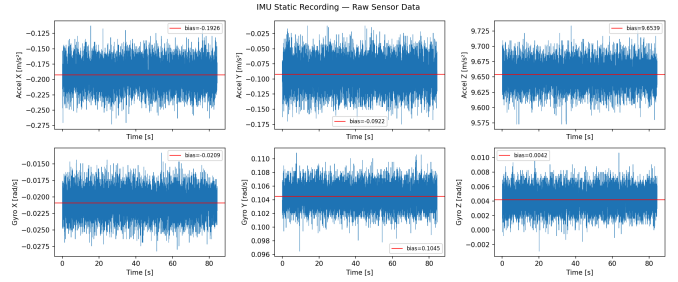


Fig. 1. IMU static recording (90 s, 100 Hz). Red lines indicate the per-axis bias. Gyro Y shows a bias of 0.105 rad/s requiring compensation.

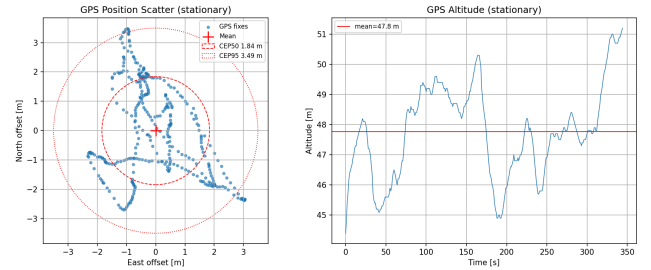


Fig. 2. Left: GPS position scatter with CEP50 (dashed) and CEP95 (dotted) circles. Right: altitude time series. CEP50 = 1.84 m, CEP95 = 3.49 m.

III. PLANNED WORK

EKF implementation (Week 4): After being moved forward in Progress Report 2, the EKF implementation returns to its original Week 4 slot. The state vector and motion model will be finalized then, using R_{GPS} and R_{IMU} derived above. Localization accuracy will be quantified by comparing GPS-only against fused estimates.

Field testing and final report (Week 5): Outdoor trajectory experiments and, if schedule allows, a Pure Pursuit or Stanley trajectory controller.

The updated project timeline is in Section IV.

IV. APPENDIX

TABLE I

MPU-6050 STATIC NOISE STATISTICS (8438 SAMPLES, 84.4 S AT 100 Hz AFTER 5 S WARM-UP). GYRO Y BIAS IN BOLD DUE TO MAGNITUDE.

Channel	Bias (mean)	Std dev (noise)
Accel X	-0.193 m/s ²	0.021 m/s ²
Accel Y	-0.092 m/s ²	0.024 m/s ²
Accel Z	+9.654 m/s ²	0.020 m/s ²
Gyro X	-0.021 rad/s	0.002 rad/s
Gyro Y	+0.105 rad/s	0.002 rad/s
Gyro Z	+0.004 rad/s	0.001 rad/s

TABLE II

PROJECT TIMELINE. WEEKS 1–3 COMPLETE.

Week	Task	Status
1	Hardware assembly, sensor interface verification	✓
2	IMU accel/gyro reads, ROS2 environment and sensor nodes	✓
3	Static sensor noise characterization, EKF parameter derivation	✓
4	EKF implementation, state vector definition, localization evaluation	
5	Field testing, trajectory controller (<i>if schedule allows</i>), final report	

REFERENCES

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